13) Momentum Gradient descent

```
import numpy as np
X = np.array([1,2,3,4,5])
Y = np.array([2,4,6,8,10])
W = 0
b = 0
lr = 0.01
epochs = 50
beta = 0.9 #momentum term
VW = 0
Vb = 0
for epoch in range(epochs):
    Y_pred = W*X + b
    dW = -2 * np.mean (X * (Y - Y_pred))
    db = -2 * np.mean (Y - Y_pred)
    VW = beta * VW + (1-beta) * dW
    Vb = beta * Vb + (1-beta) * db
    W = W - lr * VW
    b = b - lr * Vb
print ("Weight :", W)
print ("Bias :", b)
```

OUTPUT

Weight: 1.7908085636302 4

Bias : 0.42274914529 73533